Finite triangle-local search with exact rational certificates

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Abstract

We implement a finite-table search for local models in the triangle network, with independent source weights, stochastic or deterministic local responses, and exact rational verification. Joint optimization is combined with convex factor updates, independent source-symbol splitting, and perturbed searches. The benchmark compares the Elegant Joint Measurement, Renou, and Fritz families with the numerical estimates of Kriváchy et al. (2020). At the four reported noise transitions, our rational candidates have Euclidean errors below 4×10^{-8}. At noiseless Renou $u^2 = 0.85$, the certified error is 2.24×10^{-3}, below both independently reported 2020 values. The certificate checker recomputes every probability and model-to-target distance with integer arithmetic. It also verifies an explicit symmetric local witness with all-equal probability 41/144. Numerical fits and exact target witnesses are distinguished throughout: a positive search residual is an upper bound on distance to the local set.

1 Finite models and exact verification

The triangle has sources α, β, γ shared respectively by BC, CA, and AB. A finite local model consists of probability vectors $q_\alpha, q_\beta, q_\gamma$ and row-stochastic response tables A^a_{jk}, B^b_{ki}, and C^c_{ij}. Its output law is

$$P_M(a,b,c) = \sum_{i,j,k} q_\alpha(i) q_\beta(j) q_\gamma(k) A^a_{jk} B^b_{ki} C^c_{ij}. \tag{1}$$

Independence is imposed by the product of the three source laws. This is the finite form of the network-locality definition of Fritz [1]; finite cardinality bounds are established by Rosset et al. [2].

A certificate stores rational source weights, rational response rows, the rational target, and exact distances. Checking nonnegativity and row sums gives three independent random sources and valid local response kernels. Equation (1) therefore defines a triangle-local distribution. Clearing the factor denominators reduces its evaluation to integer multiplication and addition. Equality of every output entry with the target gives an exact local witness. Otherwise the same computation gives exact L1, squared Euclidean, and maximum errors, and hence a certified nearby local distribution. The Euclidean error bounds the target's distance to the unrestricted local set from above.

The implementation uses Python integer arithmetic and checks all simplexes, normalization, party permutations, and model/target hashes independently of the optimizer. It is tested software, rather than a formally verified checker. Finite alphabets and nonconvex optimization do not give a complete membership test. General Renou targets contain square roots and are stored as explicit rational approximations; their certificates refer to those stored targets.

2 Search and known local seeds

Joint L-BFGS optimization uses softmax coordinates and analytical derivatives of the full finite contraction. For fixed other factors, each individual source vector or party response table enters linearly in (1). Squared Euclidean fitting is thus a convex quadratic problem on a simplex or a product of simplexes. We solve it by projected accelerated gradient and accept only updates that decrease the evaluated objective. Inner iteration caps remain part of the numerical method. For L1 loss, each block is a sparse linear program. Deterministic descent instead minimizes over one output choice at a time; all single-row choices are evaluated

with floating-point costs and a numerical acceptance tolerance, rather than exact arithmetic. This does not give a global discrete optimum.

The hybrid method alternates joint optimization and convex polishing, then perturbs the best candidate and repeats. Output distributions rounded to seven decimal places identify duplicate fits for diagnostics. Every evaluated candidate remains eligible to improve the best value. Forward and reverse continuation and reuse of nearby fitted models reduce sensitivity to individual starts.

A local seed with larger requested alphabets is embedded by splitting each source symbol's weight among copies and pulling back its adjacent response tables. Summing over copies in (1) recovers the original model. This operation preserves source independence. Response perturbations then open additional directions for optimization. Recognized controls are used as seeds and pass through the ordinary rational certificate checker.

The supplied symmetric witness uses uniform independent sources on $S = \{(i,j) \in \{0,1,2,3\}^2 : i \neq j\}$ and

$$f(x,y) = \begin{cases} x_2, & x_2 \in \{y_1, y_2\}, \\ x_1, & \text{otherwise}, \end{cases} \qquad (A,B,C) = (f(\beta,\gamma), f(\gamma,\alpha), f(\alpha,\beta)).$$

Enumeration of the 12^3 source triples gives counts 123, 19, and 23 for each output triple with one, two, or three distinct labels, respectively. Consequently $P(A = B = C) = 4(123)/1728 = 41/144 > 1/4$, with full party and joint-outcome symmetry. This supplies an exact witness relevant to Bäumer et al. [4]. The restricted-class enumeration and a revision-pinned link to the Lean source are retained in the supporting witness note; this benchmark runs the independent integer checker. The four-symbol token/parity seed for Renou's $u^2 = 1/2$ uses independent token bits and independent edge parity bits and reproduces that endpoint exactly. Fritz's CHSH-local embeddings and uniform noise are additional exact controls.

3 Published targets and benchmark comparison

Kriváchy et al. [3] use Euclidean distance in Eq. (3). Their Eq. (4) approximates a noisy-family distance by zero below a fitted v_* and by $\|P(v) - P(v_*)\|_2 \sin\theta$ above it. Figure 4 and the discussion on page 5 report EJM source/detector pairs $(v_*, \theta) \simeq (0.80, 50°)$ and $(0.86, 60°)$. Figure 5 and page 5 report Renou pairs $(0.89, 6°)$ and $(0.91, 6°)$ at $u^2 = 0.85$. These are heuristic numerical estimates; the paper explicitly describes the Renou estimates as cruder and sensitive to local optima and variation between independent runs. Our reference columns evaluate that equation; they are not digitized training data. Gitton and Renner give a computer-assisted nonclassicality proof for the ideal EJM [6], Section 6.2, independently of the present search. Their Appendix C.2 uses $vP_{\text{pure}} + (1-v)/64$ with equality-pattern entries $(1/8, 0, 1/48)$ for P_{pure}; the ideal EJM occurs at $v = 3/4$. That parameter is distinct from our independent source and detector visibilities.

For EJM, the probabilities of triples with one, two, and three distinct labels are the entries of

$$\frac{1}{256}\left(4 + 9v + 9v^2 + 3v^3,\ 4 + v - 3v^2 - v^3,\ 4 - 3v + 3v^2 + v^3\right)$$

under independent source noise, and

$$\frac{1}{256}\left(4 + 9v^2 + 12v^3,\ 4 + v^2 - 4v^3,\ 4 - 3v^2 + 4v^3\right)$$

under independent detector noise. These distinct physical families are not replaced by a global white-noise mixture. Renou's family [5] uses three $|\phi^+\rangle$ sources and the measurement basis $|01\rangle, |10\rangle, u|00\rangle + \sqrt{1-u^2}|11\rangle, \sqrt{1-u^2}|00\rangle - u|11\rangle$ in cyclic order. Source noise uses a six-qubit Born-rule contraction; detector noise applies independent uniform replacement channels. The generators were checked against the 2020 authors' revision-pinned target code; the largest entry discrepancy in 24 initial checks was 4.21×10^{-13}. A further 60 Renou source/detector checks, including the high-visibility fits, agreed within 6.74×10^{-13} after one spelling correction in the reference detector branch, applied in memory.

Table 1 compares the first sweep with the final selected rational models at the four reported noise transitions. Small certified distances establish nearby local models; none of these small residuals is substituted for exact target membership. Certificates for Renou refer to the stored rational targets. Independent 70-digit evaluations of the authors' source-noise polynomial and the detector channel found Euclidean target-approximation errors of 5.23×10^{-13} at source $v = 0.89$, 2.10×10^{-14} at detector $v = 0.91$, and 2.78×10^{-14} at noiseless $u^2 = 0.85$. These are numerical cross-checks of target generation, separate from exact certificate arithmetic. The full scans, including both directions and unsuccessful fits, are retained in CSV form. Current-model decimal distances are rounded from exact rational squared-distance certificates.

Family	v	First-sweep d	Current certified d	Certified L1
EJM source	0.80	5.38×10^{-3}	5.77×10^{-11}	3.7×10^{-10}
EJM detector	0.86	0.01088	1.51×10^{-9}	8.68×10^{-9}
Renou source	0.89	1.07×10^{-3}	3.27×10^{-8}	2.05×10^{-7}
Renou detector	0.91	8.29×10^{-4}	5.91×10^{-9}	3.87×10^{-8}

Table 1: Distance upper bounds at the reported 2020 transitions.

The noiseless Renou scan includes the known local endpoints and the difficult region near $u^2 \simeq 0.785$ (Table 2). At $u^2 = 0.85$, the final certified distance is 2.24×10^{-3}; the 2020 paper separately reports about 0.0034 and 0.0055 in two runs (Figure 5, page 5). Our candidate is below both reported values. The ideal EJM error 0.04834 is about 0.37% above the source Eq. (4) estimate 0.04816; the detector estimate is 0.05268; these approximate curves do not supply the authors' raw saved models. These comparisons describe candidate quality, not lower bounds on the true distances. Blind L-BFGS still misses $u^2 = 1/2$ in the component test; explicit or recognized token/parity seeding produces its exact certificate. Fritz's clipped-CHSH construction matches the reference $\max(v/\sqrt{2} - 1/2, 0)/4$, with exact rational local controls at correlator magnitudes 0, 1/4, and 1/2.

u^2	Certified Euclidean distance	Exact stored target
0.5	0	yes
0.63	8.01×10^{-3}	no
0.785	5.52×10^{-9}	no
0.85	2.24×10^{-3}	no
1	0	yes

Table 2: Selected noiseless Renou points. The decimal 0.785 is a sampled parameter, not the exact critical parameter of the original theorem.

4 Which components help

Table 3 compares configurations at EJM source visibility 0.80. Larger alphabets, deterministic descent, least squares, and the small response network are useful alternative starts, but do not by themselves recover this target in the measured trials. The strongest improvement comes from a lifted local seed, free source-weight updates, and perturbed hybrid searches. The symmetric families are restricted constructions; their persistent residuals are not nonlocality certificates. The neural component has two hidden ReLU layers of width 24 on finite source grids and uses NumPy Adam. It is distinct from the 2020 EJM architecture (depth 5, width 30, Monte Carlo batches around 8,000; page 6 of [3]).

Configuration	Certified distance	Search seconds
Blind L-BFGS, 12 symbols	0.01985	4.6
Blind L-BFGS, 24 symbols	0.01211	32.0
Least squares, 12 symbols	8.06×10^{-3}	3.2
Deterministic descent, 24 symbols	8.35×10^{-3}	11.2
Pair symmetry, 12 symbols	5.38×10^{-3}	0.5
Pair symmetry, 24 symbols	5.38×10^{-3}	2.8
NumPy response network, 24 symbols	0.01707	23.8
Lifted witness + L-BFGS	5.43×10^{-3}	3.8
Then convex polish, fixed weights	1.63×10^{-3}	5.8
Then convex polish, free weights	4.07×10^{-4}	8.2
Seeded hybrid + two perturbations	5.06×10^{-10}	28.0

Table 3: Component comparisons use the same pseudorandom seed, with recorded method-specific iteration and restart caps. They are descriptive comparisons, not equal-budget statistical estimates. Polishing rows start from the same seeded L-BFGS model and report their additional runtime.

5 Reproduction

The NumPy/SciPy package provides JSON/CSV input, a command line, MIT licensing, and gzip-compressed rational bundles. The base sweep uses two starts, at most 1,800 joint iterations, 200 convex sweeps, and seed 20261010. Boundary refinements and component tests record their separate caps and seeds. One optimizer runs at a time with two numerical-library threads. The measured base sweep and targeted refinements took 1678.5 seconds and had largest recorded process peak RSS 104.1 MB; exploratory probes are excluded. All 50 selected bundles were independently recomputed successfully. These timings are not a hardware-matched comparison with the original neural-network training.

The repository README gives installation, complete reproduction commands, certificate schemas, and target arithmetic conventions. The tests exercise analytical Jacobians, convex descent, seed lifting, exact controls, malformed and altered certificates, and command-line round trips.

References

[1] T. Fritz, Beyond Bell's theorem: correlation scenarios, *New J. Phys.* **14**, 103001 (2012), doi:10.1088/1367-2630/14/10/103001.

[2] D. Rosset, N. Gisin, and E. Wolfe, Universal bound on the cardinality of local hidden variables in networks, *Quantum Inf. Comput.* **18**, 910–926 (2018), doi:10.26421/QIC18.11-12-2.

[3] T. Kriváchy, Y. Cai, D. Cavalcanti, A. Tavakoli, N. Gisin, and N. Brunner, A neural network oracle for quantum nonlocality problems in networks, *npj Quantum Inf.* **6**, 70 (2020), doi:10.1038/s41534-020-00305-x.

[4] E. Bäumer, V. Gitton, T. Kriváchy, N. Gisin, and R. Renner, Exploring the local landscape in the triangle network, *Phys. Rev. A* **111**, 052453 (2025), doi:10.1103/PhysRevA.111.052453.

[5] M.-O. Renou, E. Bäumer, S. Boreiri, N. Brunner, N. Gisin, and S. Beigi, Genuine Quantum Nonlocality in the Triangle Network, *Phys. Rev. Lett.* **123**, 140401 (2019), doi:10.1103/PhysRevLett.123.140401.

[6] V. Gitton and R. Renner, The Elegant Joint Measurement is Non-Classical in the Triangle Network, arXiv:2510.15143v1 (2025).